

CLOUD COMPUTING

INDIVIDUAL REAL-TIME PROJECT ASSIGNMENT

Disaster Recovery, Backup & High-Availability Cloud Platform

Hi Sujith,

Your second individual Cloud Computing real-time project assignment is provided below. Kindly complete the project independently and submit the implementation, documentation and testing evidence within **15–20 days** from the date of assignment.

Project Title: Multi-Region Disaster Recovery & Business Continuity Platform

Design and implement a cloud-based disaster recovery solution for a business-critical web application. The system must continue operating during infrastructure failure and provide a practical strategy for backup, recovery, monitoring, security and controlled failover.

Business Scenario

A financial-services application receives continuous customer requests and stores important transactional data. The organization cannot depend on a single server or single availability zone. You are required to design a resilient cloud architecture that minimizes downtime and data loss during failures.

Core Technical Requirements

Component	Expected Work
High Availability	Deploy redundant application resources across availability zones where supported.
Disaster Recovery	Design a secondary recovery environment and define a failover process.
Database	Use a managed database with backup, replication or another suitable recovery mechanism.
Storage	Protect critical files using object-storage versioning, lifecycle and backup controls.
Networking	Design secure public/private network segments, routing and firewall rules.
Security	Implement IAM least privilege, encryption and secure secrets handling.
Monitoring	Create dashboards, health checks, logs and failure alerts.
Automation	Automate deployment and recovery steps using scripts or infrastructure-as-code where possible.
Testing	Perform a controlled failure simulation and document recovery results.

Technical Questions / Deliverables

1. Draw the complete high-availability and disaster-recovery architecture and explain every component.
2. What are the single points of failure in your initial design, and how will you eliminate or reduce them?
3. Define the Recovery Point Objective (RPO) and Recovery Time Objective (RTO) for the application and justify your targets.
4. Explain how application traffic will be redirected during a regional or infrastructure failure.
5. How will you protect the database against accidental deletion, corruption and infrastructure failure?
6. Demonstrate a backup strategy and explain retention, restoration and verification of backups.
7. How will you secure the recovery environment so that it does not become a weaker security boundary?
8. Design monitoring and alert rules that can identify service failure, high latency, storage problems and backup failures.
9. Perform a controlled failure test. Record the failure, detection time, recovery steps and final recovery time.

10. What data-consistency problems could occur during failover, and how would you mitigate them?
11. Explain how IAM roles, encryption keys and secrets would be handled across primary and recovery environments.
12. Design a CI/CD process that can deploy the application consistently to both environments.
13. Estimate the major cloud costs of keeping a recovery environment available and propose cost-saving alternatives.
14. What would happen if the primary database became unavailable during a high-traffic period? Describe the sequence of actions.
15. Prepare a final disaster-recovery runbook that another engineer could follow without needing your assistance.

Submission Package

- Source-code repository with README and project structure.
- Cloud architecture diagram showing primary and recovery environments.
- Deployment/configuration documentation without passwords, access keys or private credentials.
- Backup and restoration evidence, monitoring screenshots/logs and failure-test results.
- RPO/RTO explanation and disaster-recovery runbook.
- Final technical report and short project demonstration if requested.

Evaluation & Performance

Area	Weight
Cloud Architecture & Resilience	20%
Implementation & Deployment	20%
Backup & Disaster Recovery	20%
Security & IAM	15%
Monitoring & Testing	15%
Documentation & Presentation	10%

Performance-linked stipend note: Following submission, the project will undergo technical performance evaluation. As per the communicated program terms, a performance score of **50% and above** will be considered for the applicable stipend criteria, with the stated range of **■15,000–■20,000**, subject to program rules, verification and approval. This assignment document is not a guarantee of stipend payment.

Please complete the project within the assigned timeline and update the project coordinator once the final submission is ready for evaluation.

Regards,

Training & Internship Project Evaluation Team
Cloud Computing Program